

HeartIQ — Intelligent Heart Disease Prediction System

1. Introduction

HeartIQ is a data-driven system designed to predict the likelihood of heart disease using machine learning algorithms. It analyzes patient health data such as age, cholesterol, blood pressure, and lifestyle habits to identify potential risks early. The system helps doctors and patients make informed decisions, promoting preventive healthcare through predictive analytics.

2. Required Tools

- Python 3.10+ — core language for development
- Pandas and NumPy — for data preprocessing and analysis
- Scikit-learn — to build and train machine learning models
- Matplotlib / Seaborn — for visualizing health data trends
- Flask or Streamlit — to build a simple user interface
- SQL / CSV files — for data storage
- Jupyter Notebook / VS Code — for model experimentation

3. How to Make the Project

- 1 Step 1: Collect and clean patient data (age, cholesterol, blood pressure, etc.).
- 2 Step 2: Perform data analysis to find correlations between features and heart disease.
- 3 Step 3: Split the dataset into training and testing sets.
- 4 Step 4: Train classification models like Logistic Regression, Random Forest, or SVM.
- 5 Step 5: Evaluate the model using accuracy, precision, recall, and ROC curve.
- 6 Step 6: Visualize important health factors influencing prediction outcomes.
- 7 Step 7: Build an interactive web app to allow users to input data and get predictions.

4. Problem Solved

HeartIQ addresses the critical issue of late detection of heart diseases. Many individuals remain unaware of their cardiac risks until symptoms become severe. By using predictive machine learning techniques, HeartIQ provides early warnings and personalized health insights. This enables preventive action, reduces hospitalizations, and promotes overall cardiac wellness.